Supersession Data
(See Section 6)

MILITARY SPECIFICATION

HEAD COIL ASSEMBLY

This specification is approved for use by the Naval Air System Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the AIM-9M Head Coil Assembly used in the Guidance Control Section (GCS) WGU-4/B or WGU-4A/B and is referred to herein as the "head coil."

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-591	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

MILITARY

MIL-P-116	Preservation-Packaging, Methods of.
MIL-T-18303	Test Procedures, Preproduction Acceptance, and Life for Aircraft Electronic Equipment, Format for.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.
FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications of the issue specified in the solicitation, form a part of this specification to the extent specified herein.

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DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

639AS3010	Head Coil Potted Assembly.
639AS3114	Dome, Housing Assembly.
639AS3180	Gyro, Optical Assembly.
2412395	Sheet, Technical Frequency Specifications.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

AD-1322	Reliability Program Provisions for Production of Sidewinder AIM-9.
WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, IL 60606.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI S1.4-1971 Sound Level Meters.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 Item description. The head coil is an encapsulated assembly consisting of various coils wound on a molded bobbin. The head coil contains two sets of gyro drive coils, two sets of reference coils, precession coils, caging and cage cancel coils, and acquisition precession coil.

3.2 Conformance to documents. The head coil shall conform to the requirements specified herein and to Drawing 639AS3010.

3.3 Design and construction. The design, construction, materials, processes, and finish shall be such that the head coil shall meet all requirements as specified herein. Acceptance or approval of any constituent material shall not be construed as acceptance of the finished product. Certification, accompanied by objective quality evidence, in accordance with MIL-STD-109, shall be readily available for inspection by the Government, certifying that the materials and components used comply with the applicable specification for parts and components which require such data.

3.4 Performance requirements.

3.4.1 Cage cancel adjustment. The cage cancel resistance shall be determined as specified herein and shall be permanently marked on the head coil as specified on Drawing 639AS3010.

3.4.2 Electrical characteristics. The head coil shall meet the requirements for induced voltage, direct current (dc) resistance, and reference voltage phase as specified in Table I.

3.4.3 Boresight. The angle between the gyro spin axis and the head coil mounting surface (see Datum C of Drawing 639AS3010) shall be 90 degrees ±0 degrees 15 minutes.

3.4.4 Insulation resistance. The insulation resistance shall be 10,000 megohm (Mohm) minimum between nonconnected coils and between any coil and the metallic insert of the gyro support molded assembly.

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TABLE I. Electrical requirements.

Coil designation	Header pin no.		Induced voltage 1/ (Vtrms)	DC resistance at 20°C (ohms)	Phase (degrees)
	From	To			
Right/left reference voltage	18	3	1.05 ±8%	10.8 ±22%	
Up/down reference voltage	9	3	1.05 ±8%	10.8 ±22%	
Reference voltage phase Up/down leads Right/left	18	9			90 ±4
Cage cancel coil	7	3		716 ±12%	
Cage coil	7	6		617 ±12%	
Precession coil	5	4		13 ±10%	
Motor drive 1	11	10	4.2 Min 2/	15.2 ±10%	
Motor drive 2	1	2	4.2 Min 2/	15.2 ±10%	
Acquisition precession coil	15	3		160 ±10%	

1/ These are open circuit measurements for induced voltage.

2/ Motor drive 2 induced voltage shall equal motor drive 1 induced voltage ±0.10 Vtrms.

3.5 Environmental requirements. The head coil shall withstand the following environments and shall show no evidence of any physical or performance degradation.

3.5.1 Thermal shock. The head coil shall withstand the thermal shock conditions of MIL-STD-202, Method 107, with the exceptions specified herein, and shall thereafter meet the requirements specified herein.

3.5.2 Humidity. The head coil shall withstand the humidity conditions of MIL-STD-202, Method 103, Test Condition B, and shall thereafter meet the requirements specified herein.

3.5.3 Hot storage. The head coil shall withstand exposure to high temperatures of 85 ±3 degrees Celsius (°C) for a period of 96 hours minimum and shall thereafter meet the requirements specified herein.

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3.5.4 Cold storage. The head coil shall withstand exposure to cold temperatures of -65 ±3°C for a period of 96 hours minimum and shall thereafter meet the requirements specified herein.

3.5.5 Vibration. The head coil shall withstand the vibration conditions of MIL-STD-202, Method 204, Test Condition B, with the exceptions specified herein, and shall thereafter meet the requirements specified herein.

3.5.6 Shock. The head coil shall withstand the shock conditions of MIL-STD-202, Method 213, with the exceptions specified herein, and shall thereafter meet the requirements specified herein.

3.5.7 Life. The head coil shall withstand the life test conditions of MIL-STD-202, Method 108, Test Condition C, with the exceptions specified herein. The head coil shall exhibit no evidence of physical deterioration, and shall thereafter meet the electrical requirements specified herein.

3.6 Standards of manufacture.

3.6.1 Special working conditions. The facilities used in the manufacture, testing, and inspection of the head coil shall be in accordance with the provisions of Figure 1.

3.6.2 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536.

3.6.3 Marking. Each head coil shall be marked in accordance with MIL-STD-130, with the part number and manufacturer's name or symbol. Marking shall include the value of the cage cancel resistor in the location specified on Drawing 639AS3010.

3.7 First article. When specified in the contract or purchase order, a sample of head coils shall be subjected to first article inspection (see 4.2 and 6.2.3).

3.8 Appearance. When the head coil is tested as specified herein, the encapsulant shall remain as specified in 3.9.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. Fabrication and finishing shall be accomplished in a manner that will ensure freedom from cracks, chips, blisters, wrinkles, and voids in the casting material or any surfaces. Any patching in or on a cast surface shall have been accomplished with compatible materials. All flashing and spillage shall have been removed and the gyro-mount-insert-mounting surfaces shall be free from paint, grease,

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Type of Work	Cleanliness Note 1	Lighting Foot Candles Note 2	Air Temperature Note 3	Relative Humidity Note 4	Dust Control Note 5	Ventilation or Exhaust Note 6	Noise Note 7 dbA	Habitat
Machine Shop								
Tolerance to 0.01	Class D	70	60-100°F	U	Class D	Note 6	90 dbA	Note 8
Tolerance to 0.001	Class C	100	65-95°F	U	Class C	Note 6	80 dbA	Note 8
Tolerance to 0.0001	Class B	200	65-85°F	70% Max.	Class C	Note 6	80 dbA	Note 8
Tolerance finer than 0.0001	Class BA	500	67-75°F	60% Max.	Class B	Note 6	75 dbA	Note 9
Propellant Chemical and Potting Areas	Class B	100	65-95°F	60% Max.	Class B	Note 6	80 dbA	Note 9
Plating and Heat Treating	Class C	70	55-105°F	U	Class C	Note 6	90 dbA	Note 8
Electronic Assembly	Class B	100	65-95°F	30-70%	Class B	Note 6	75 dbA	Note 9
Mechanical Assembly								
Tolerance to 0.05	Class C	100	65-95°F	U	Class C	Note 6	75 dbA	Note 8
Tolerance to 0.001	Class B	100	65-95°F	U	Class C	Note 6	75 dbA	Note 8
Tolerance to 0.0001	Class BA	200	65-85°F	70% Max.	Class B	Not Req	70 dbA	Note 9
Tolerance finer than 0.0001	Class A	500	65-75°F	30-50%	Class A	Not Req	70 dbA	Note 9
Inspection Stations	Equal to product inspected							

1. Cleanliness Definitions

   a. Class D - - Daily Cleanup: Removal of scrap, clean up all spilled oil, etc.

   b. Class C - - Prompt Cleanup: Scrap, oil and residue shall not be allowed to accumulate.

   c. Class B - - Prompt Cleanup: Oil, residue, spilled chemicals removed immediately. Floors, walls and work area shall have hard, grease resistant, easily cleaned surfaces. Cleanup of equipment and area shall be accomplished after each 24 hours of operation.

   d. Class BA - - Oil, residue, spilled chemicals, and any foreign material which might develop in the area shall be cleaned immediately. Floors, walls, furniture and work areas shall have hard, grease resistant, easily cleaned surfaces. Materials to be worked on shall be cleaned of all foreign material prior to acceptance into this area. Flushing or cutting fluids shall be filtered to remove contamination above 30.0 microns. Chips, particles and dust generated during any and all mechanical operation shall be removed during the operation by vacuum. Clean lint-free smocks are required for operators and inspectors. Complete cleaning and wipe down of equipment, tools, fixtures and the area shall be accomplished at the end of each 8 hours of operation or each 24 hour period.

   e. Class A - - Cleanliness controlled in accordance with FED-STD-209, Class 100,000.

2. Lighting

   Indicated values are minimum light intensity values at the work station. Supplemental lighting shall be used when necessary to improve precision and minimize operator fatigue, but brightness ratios within the operators' field-of-view shall not exceed 10 to 1.

3. Air Temperature

   Designated temperature limits are the daily average temperature measurements taken in proximity of the work stations. Ambient temperature shall be between 68°F and 85°F.

4. Relative Humidity

   Designated relative humidity shall be as measured at room ambient temperature. "U" indicates uncontrolled relative humidity.

5. Dust Control Definitions

   a. Class D - - No dust control required.

   b. Class C - - Outside air shall be filtered to remove dust particles. Type of filter is unspecified

   c. Class B - - Outside and recirculated air shall be filtered to remove dust particles. Filter rating shall be 10 micron maximum.

   d. Class A - - Dust control shall be in accordance with FED-STD-209, Class 100,000.

6. Ventilation or Exhaust

   Forced ventilation or exhaust shall be provided whenever required to minimize operator fatigue.

7. Noise

   Noise is defined as the average sound level existing at the work station during normal operation when measured with a standard sound-level meter as specified in ANSI S1.4-1971 "A" weighted. Work stations for control of automatic equipment-operations, which requires operator loading and unloading only, may exceed the noise limits specified by 10 dbA but not to exceed 90 dbA. Inspection and test work stations shall not exceed the values specified. Work areas for mechanical assembly of tolerances finer than 0.001 which utilize laminar flow benches may exceed by 5 dbA the value specified.

8. Habitat

   No eating, drinking or personal grooming is allowed in these work areas. Drinking fountains are permissible.

9. Habitat

   No eating, drinking, smoking or personal grooming is allowed in these work areas. Drinking fountains are permissible.

FIGURE 1. Minimum standards for working environments.

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potting material, and any other foreign contaminant. The header shall contain no damaged or bent pins. All threaded holes in the gyro mount insert shall be free from foreign material. The encapsulant shall not corrode any metals being coated or in contact with potting compounds.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection and testing requirements are classified as follows:

a. First article inspection (see 4.2).

b. Quality conformance inspection (see 4.4).

(1) Acceptance tests.

(2) Environmental tests.

4.2 First article inspections. When specified in the contract or purchase order, first article inspections shall be performed by the contractor after award of contract and prior to production (see 6.2.3). As specified in the contract or purchase order, first article inspections shall be performed on six head coils which have been produced with the equipment, material, and procedures to be used in production (see 6.2.3). Any production prior to approval of the first article shall be at the risk of the contractor.

4.2.1 First article inspection sequence. First article inspections shall consist of the Groups A, B, and C tests of Table II, in the sequence shown. Each of the Group C tests shall be followed by the Group B tests as specified herein.

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TABLE II. First article and quality conformance tests.

Inspection	Requirement paragraph	Method paragraph
Group A		
Cage cancel adjustment	3.4.1	4.5.2
Group B		
Visual	3.2, 3.3, 3.6	4.5.1
Electrical characteristics	3.4.2	4.5.3
Boresight	3.4.3	4.5.4
Insulation resistance	3.4.4	4.5.5
Group C		
Thermal shock	3.5.1	4.6.1a, 4.6.1b
Humidity	3.5.2	4.6.2
Hot storage	3.5.3	4.6.3
Cold storage	3.5.4	4.6.4
Vibration	3.5.5	4.6.5
Shock	3.5.6	4.6.6
Life	3.5.7	4.6.7
Group D		
Thermal shock	3.5.1	4.6.1a.
Hot storage	3.5.3	4.6.3
Cold storage	3.5.4	4.6.4

4.2.2 First article verification. Unless otherwise specified in the contract or purchase order (see 6.2.3), the contractor shall be required to perform first article verification (resubmission for first article testing) during the production phase under any of the following conditions:

a. The contractor has modified his product through a change of raw materials, processes, production procedures, or methods. It shall be the responsibility of the contractor to notify the

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procuring activity prior to the incorporation of any such changes and to provide quantitative evidence of the effect of such changes on the performance or characteristics of the product. Requirement for such tests will be based on an evaluation of the evidence supplied.

b. Where there is evidence that the quality level of the product has not been maintained. This evidence may be in the form of accumulated failure reports of the product, system failures attributable to the product, or failure of the product to pass any of the tests that may be conducted by or for the procuring activity.

c. Any change in design or documentation by the procuring activity.

d. When the head coil is out of production for any period exceeding 18 months.

4.2.3 Acceptance criteria. Failure of any head coil to pass any of the tests specified herein shall reject first article sample.

4.3 General test conditions and provisions. Unless otherwise specified herein, all tests, inspections, and measurements shall be performed in accordance with the following conditions and provisions.

a. Temperature: 18 to 35°C.

b. Altitude: Ground elevation of test facility.

c. Vibration: Condition of test facility.

d. Humidity: Ambient (not to exceed 95 percent relative).

4.3.1 Test equipment and facilities. Test equipment and facilities shall be provided by the contractor, and shall be of sufficient accuracy, quality, and quantity to permit performance of the inspections, tests, and measurements specified herein. Unless otherwise specified in the contract or purchase order (see 6.2.1), the limit of error in measurement apparatus shall not exceed one-fifth of the tolerance of the specified variable to be measured (see 6.3).

4.3.2 Test station certification. When specified in the contract or purchase order, all test stations used for final acceptance of head coils shall require certification by the procuring activity (see 6.2.2).

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4.3.3 Calibration. All test and measurement equipment shall be calibrated and maintained in accordance with the requirements of MIL-STD-45662. The contractor shall certify that all test equipment has been so calibrated before testing begins. When specified in the contract or purchase order, certification to this effect shall accompany all reports of first article and quality conformance inspection (see 6.2.2).

4.3.4 Test procedures. When specified in the contract or purchase order, prior to performance of any tests specified herein, detailed test procedures conforming to the applicable requirements of this specification and MIL-T-18303 shall be prepared for the procuring activity for approval (see 6.2.2). All testing shall be performed in accordance with the approved procedures, and deviation therefrom or changes thereto shall require prior approval of the procuring activity.

4 3.5 Test reports. When specified in the contract or purchase order, reports of the results of first article testing shall be prepared in accordance with MIL-STD-831 for the procuring activity for review prior to fabrication of production hardware. Reports shall contain the quantitative results of each test, inspection or measurement specified (see 6.2.2).

4.3.6 Test plans. When specified in the contract or purchase order, the contractor shall prepare a detailed test plan (see 6.2.2) for first article testing for procuring activity approval prior to the commencement of testing.

4.3.7 Action in the event of a failure. Failure analysis and reporting shall be immediately initiated in the event of a failure of a head coil while undergoing the tests of 4.2. Procuring activity concurrence shall be required for any corrective design changes.

4.3.8 Failure reporting. When specified in the contract or purchase order, head coil failures experienced during first article tests or quality conformance tests shall be documented in accordance with AD-1322. When specified in the contract or purchase order, head coil production quantity and failure report quantity shall be prepared as a monthly summary report (see 6.2.2).

4.4 Quality conformance tests. The quality conformance tests shall consist of acceptance and environmental tests.

4.4.1 Acceptance tests (100 percent screening). Each head coil shall be subjected to the Groups A and B tests of Table II. Failure of any head coil to meet any of these requirements shall cause rejection of that head coil.

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4.4.2 Environmental sampling. Each head coil in the sample shall be subjected to the Group D tests of Table II. Each of the Group D tests shall be followed by the Group B tests as specified herein. Sampling shall be in accordance with MIL-STD-105, Inspection Level I, Acceptable Quality Level (AQL) of 1.5 percent defective.

4.4.2.1 Lot description. An inspection lot shall be as specified in MIL-STD-105, and shall consist of only those products which have passed the acceptance tests specified in 4.4.1.

4.5 Test methods.

4.5.1 Visual and mechanical inspection. The head coil shall be examined for conformance to workmanship, materials, dimensions, finish, and identification markings. The head coil shall meet the requirements of 3.2, 3.3, and 3.6.

4.5.2 Cage cancel resistor. The head coil shall be operated using the cage cancel test circuit of Drawing 639AS3010. The input signal to the precession coil (pins 4 and 5) shall be a sine wave of 1.0 ±0.1 volts true root-mean-square (Vtrms) at F_1 ±1.0 Hz (see Drawing 2412395). Maximum input signal distortion shall be less than 5 percent. A gyro assembly (see Drawing 639AS3180 or equal approved by the procuring activity), with a fixed axis (gimbal locked) shall be mounted in the head coil assembly. The test circuit variable resistor shall be adjusted to obtain a minimum voltage across the cage and cage cancel coils (pin 6 to pin 3). The resistance value shall be permanently marked on the head coil as specified in 3.6.3.

4.5.3 Electrical characteristics. The head coil shall meet the requirements for each of the electrical characteristics listed in Table I. The induced voltages are open circuit measurements and shall be determined using a gyro assembly (see Drawing 639AS3180 or equal approved by the procuring activity) mounted in the head coil assembly and rotated about the gyro spin axis at F_1 ±1.0 Hz (see Drawing 2412395). The reference voltage phase shall be measured under the same conditions.

4.5.4 Boresight. A gyro assembly (see Drawing 639AS3180 or equal approved by the procuring activity) shall be mounted in the head coil assembly, rotated at F_1 ±1.0 Hz (see Drawing 2412395) and electrically caged (cage/uncage switch closed) using the boresight test circuit specified on Drawing 639AS3010. The angle between the gyro spin axis and the head coil mounting surface (see Datum C of Drawing 639AS3010) shall be measured and shall meet the requirements of 3.4.3.

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4.5.5 Insulation resistance. The insulation resistance of the head coil shall be measured in accordance with MIL-STD-202, Method 302, Test Condition B, between all nonconnected coils and between each coil and the metallic insert of the gyro support molded assembly. The electrification time shall not exceed 1 minute. The head coil shall meet the requirements of 3.4.4.

4.6 Environmental test methods.

4.6.1 Thermal shock. The head coil shall be tested in accordance with MIL-STD-202, Method 107, except the thermal shock temperatures shall be as specified in Table III starting at normal room temperature.

a. Table III constitutes one cycle and the test shall consist of a total of 5 cycles.

b. The dc resistance of each coil shall be monitored at 10 minute intervals during the temperature extremes and shall exhibit no evidence of shorted or open coils.

TABLE III. Thermal shock temperatures.

Temperature (°C)			Time (minutes)
-65	+3,	-5	60 minimum
+25	+10,	-5	5 maximum
+100	+3,	-5	60 minimum
+25	+10,	-5	5 maximum

Upon completion of the five consecutive cycles of thermal shock test, allow the head coil to return to, and remain at ambient temperature for a minimum of 2 hours. The head coil shall meet the requirements of the Group B tests of Table II.

4.6.2 Humidity. The head coil shall be tested in accordance with MIL-STD-202, Method 103, Test Condition B. Upon completion of the humidity test, allow the head coil to return to and remain at ambient temperature for a minimum of 2 hours. The head coil shall meet the requirements of the Group B tests of Table II, except the insulation resistance test potential shall be 90 percent of the values specified in 4.5.5.

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4.6.3 Hot storage. The assembly shall be placed in a temperature controlled chamber with the temperature maintained at 85 ±3°C for 96 hours. At the end of the storage period, the head coil shall be subjected to the boresight test (see 4.5.4) within 10 minutes after removal from the chamber and shall meet the requirements of 3.4.3. Upon completion of the boresight test, allow the head coil to return to and remain at ambient temperature for a minimum of 2 hours. The head coil shall meet the requirements of the Group B tests of Table II.

4.6.4 Cold storage. The test of 4.6.3 shall be repeated, except the temperature shall be -65 ±3°C.

4.6.5 Vibration. The head coil shall be tested in accordance with MIL-STD-202, Method 204, Test Condition B, except as follows:

a.

Frequencies (Hz)	Amplitude (gravity units (g))
30 to 100	7.5
100 to 1,000	10.0
1,000 to 3,000	15.0

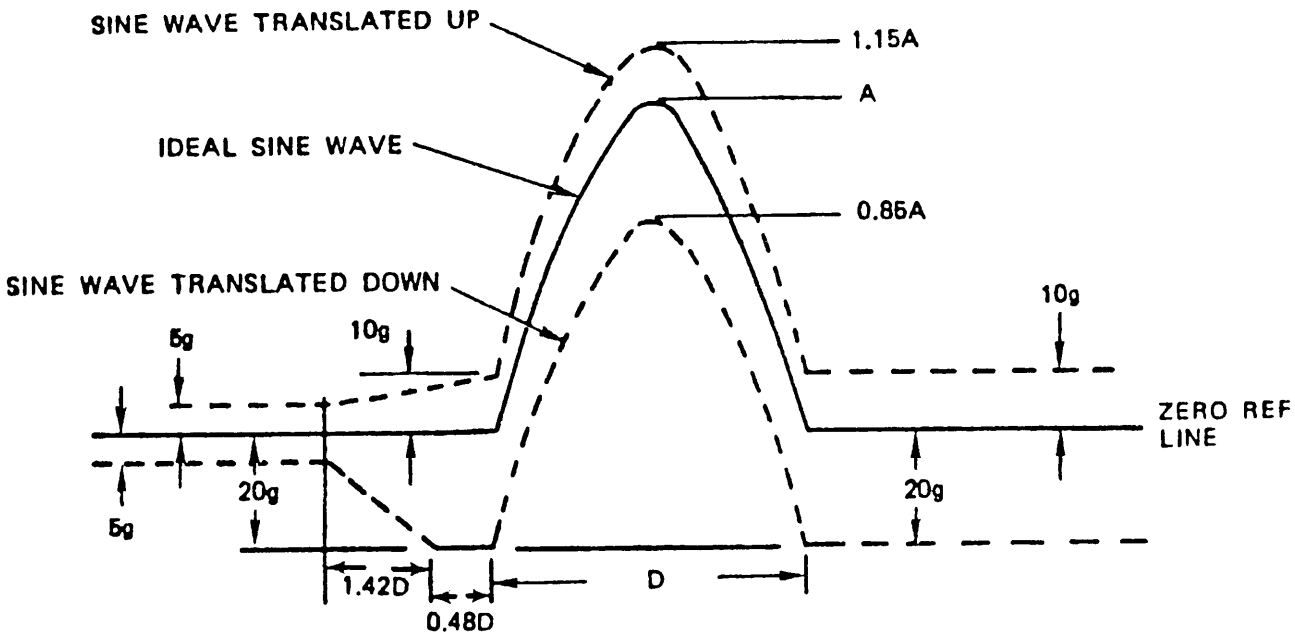
b. Sweep time shall be 5.0 ±0.5 minutes for one cycle to sweep from 30, to 3,000 Hz and return. The cycle shall be performed 12 times in each of the mutually perpendicular axes (total of 36 times) for a total period of 3 hours.

NOTE: The head coil shall be mounted within a dome housing (see Drawing 639AS-3114) which shall be secured to the vibration fixture for this test.

Upon completion of the vibration test, the head coil shall meet the requirements of the Group B tests of Table II.

4.6.6 Shock. The head coil shall be mounted in a dome housing (see Drawing 639AS3114) which shall be secured to the shock mounting fixture during the shock test. The head coil shall be subjected to shock pulses in accordance with Table IV (see Figure 2). The test unit shall be subjected to five impact shocks in each direction in the three mutually perpendicular axes (15 shocks total). Upon completion of the shock test, the head coil shall meet the requirements of the Group B tests of Table II.

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NOTE:

THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE ACCELERATION AMPLITUDE OF THE IDEAL HALF SINE PULSE IS A AND ITS DURATION IS D. THE MEASURED ACCELERATION PULSE SHALL BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES AND THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS V_i ±0.1V, WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS 2 AD/π. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4D BEFORE THE PULSE TO 0.1D AFTER THE PULSE.

FIGURE 2. Shock test.

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before use in the hydraulic system. In cases where a finer degree finer filter

TABLE IV. Shock test.

Direction	Acceleration peak, g A	Duration of acceleration, milliseconds
Longitudinal axis	47.5 ±2.5	21 ±3
Lateral axis	47.5 ±2.5	21 ±3
Vertical axis	47.5 ±2.5	21 ±3

4.6.7 Life expectancy. The head coil shall be tested in accordance with MIL-STD-202, Method 108, Test Condition C. The following details shall apply:

a. Temperature measurements shall be made at a distance not greater than 5 inches from each head coil.

b. Each head coil shall be supported by two low thermal conductivity cleats such that the minimum distance from bottom, top, and sides of the temperature chamber is 3 inches. The minimum spacing between head coils shall be 2 inches.

c. The test profile shall be a 48-hour cycle in accordance with Figure 3. The high temperature shall be +85 ±3°C and the low temperature shall be +25 ±5°C. The transition periods at the beginning and end of each cycle shall be accomplished within 55 to 90 minutes. The transition time to high temperature shall be included in the 44-hour period and the transition time to low temperature shall be included in the 4-hour period (between 44 and 48 hours).

d. The head coil shall be energized with the input levels specified in Table V for 1.5 hours, followed by 0.5 hours deenergized, successively, for the duration of the test, except that the circuits shall not be energized during the 4-hour periods of reductions to, and at 25°C as specifed in 4.6.7c.

e. Measurements immediately prior to start of the life test shall include tests for all of the electrical requirements specified in Group B of Table II. Quantitative values shall be recorded.

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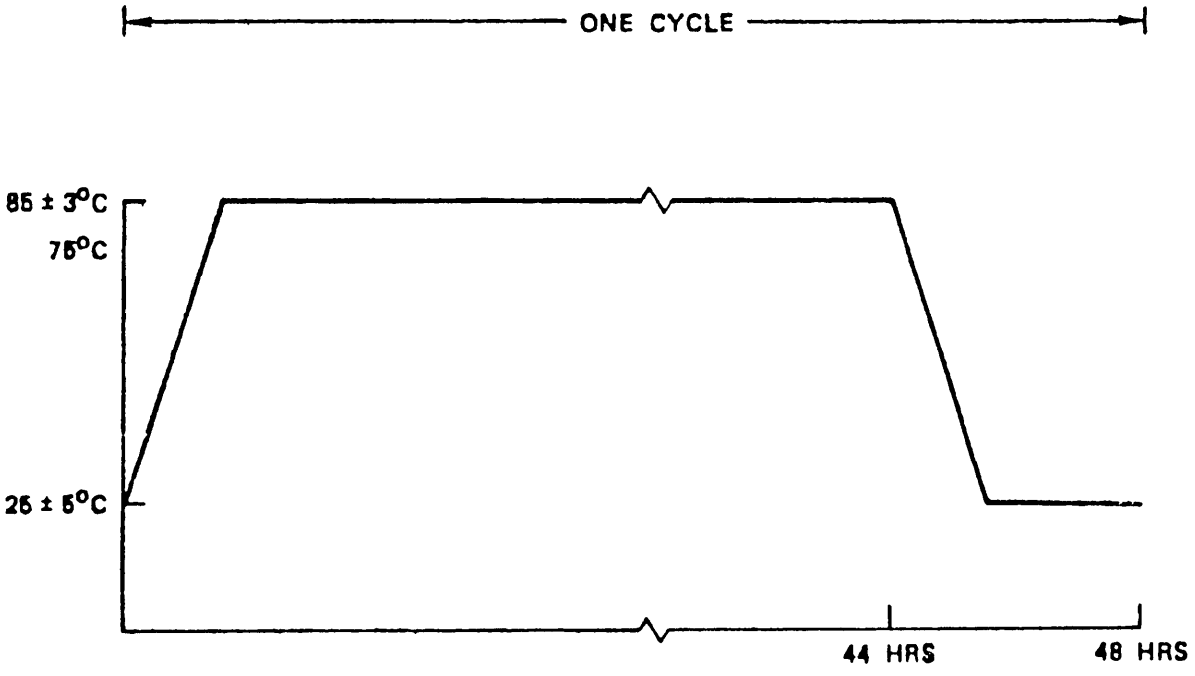


FIGURE 3. Life test profile.

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for fluid expansion or contraction. When several actuating cylinders are

f. Measurements during test shall be restricted to those measurements necessary to determine that the proper voltages are applied to the circuits, and to such other measurements as necessary to provide an indication of open or short circuit conditions of the head coil under test.

g. Measurements after the life test shall be performed when the head coil has stabilized at ambient conditions in accordance with 4.3. The measurements shall be the Group B tests of Table II. Quantitative values shall be recorded for all tests and shall be within ±5 percent of the initial values measured prior to life tests.

TABLE V. Life test input levels.

Coil designation	Header pin no.		Input level sine wave at F_1 ±1 Hz or equivalent dc
	From	To	
Motor drive 1	11	10	100 mA trms
Motor drive 2	1	2	100 mA trms
Precession coil	5	4	100 mA trms
Right/left reference coil	18	3	1 mA trms
Up/down reference coil	9	3	1 mA trms
Cage coils	6	3	1 mA trms

5. PACKAGING.

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level A or C, as specified in the contract or purchase order (see 6.2.1).

5.1.1 Level A packaging. Each head coil shall be individually protected and unit packaged in accordance with MIL-P-116, Method II(d) or Method II(f), with the following exceptions:

a. Do not use contact preservative.

b. Desiccant bag shall be secured to aft end of head coil to prevent any possibility of damage to painted surface C and area D (see applicable drawings).

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5.1.2 Level C packaging. The head coil shall be packaged to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Packages shall comply with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level A or C, as specified in the contract or purchase order (see 6.2.1).

5.2.1 Level A. The head coil, packaged as specified in 5.1 shall be packed in accordance with the requirements specified in PPP-B-591, PPP-B-601, PPP-B-621, or PPP-B-636, as applicable.

5.2.2 Level C. The head coils, packaged in accordance with 5.1, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.3.2 Normal marking. In addition to the markings required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES.

6.1 Intended use. The head coil is intended for use in the Guidance Control Section of an air-launched guided missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

c. Test equipment accuracies, if other than as specified in 4.3.1.

d. Level of preservation and packaging (see 5.1).

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e. Level of packing (see 5.2).

f. Special marking (see 5.3.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance wtih the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulations (DAR) 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirements	Applicable DID	Options
4.3.2	Certification	UDI-A-23264B	---
4.3.3	Calibration	DI-R-7064	---
4.3.4	Test Procedures	UDI-T-21347	---
4.3.5	Test Reports	DI-T-2072	---
4.3.6	Test Plans	DI-T-5204	---
4.3.8	Failure Reporting	UDI-R-21139	---

(DIDs related to this specification and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Volume II, Acquisition Management Systems and Data Requirements Control List (AMSDL). Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of six head coils as specified in the contract or purchase order (see 3.7 and 4.2). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.3 Definition.

6.3.1 Tolerance. Tolerance is defined as the permissible deviation from a specified value. For the purpose of determining test equipment

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accuracies required by this specification, when a tolerance is specified by different positive and negative excursions from the specified value, the larger value is used. Example:

a. Required temperature is 100 +3, -2°C.

b. The larger tolerance value is +3, corresponding to 3 percent, in this case.

c. The required test equipment accuracy would be ±0.6 percent.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Supersession data.

a. MIL-H-85286(AS) dated 10 January 1980.

b. MIL-H-85286(AS), Amendment 1 dated 16 October 1980.

c. MIL-H-85286(AS), ECP F1031R1 (NOR F1031.1A) dated 14 January 1983.

Preparing activity:
Navy (AS)
Project 1420-N267)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-85286A	HEAD COIL ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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